BIG SYNTAX DOESN'T WANT YOU TO KNOW

MAL is universal? Hardly.

The Menzerath–Altmann law on all 180 UD treebanks

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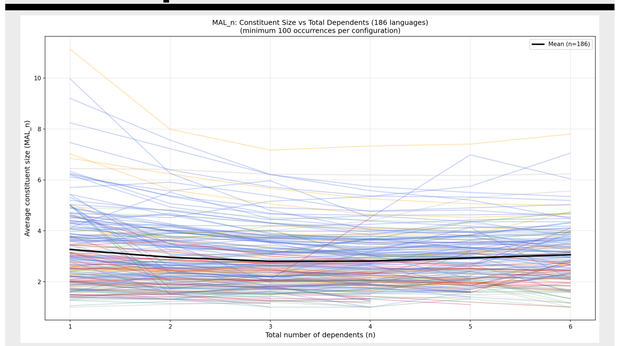
LINGUISTS HATE THIS TRICK

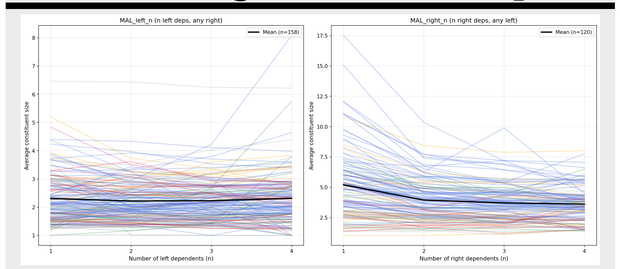
Got more dependents? Just make them smaller!



In **most languages, yes.** The average size of a verb's dependents drops as the dependent count **n** grows — exactly what Menzerath-Altmann predicts. **Each thin line = one of 186 languages**; the bold line is the cross-linguistic mean.

THE TRUTH THEY ARE HIDING FROM YOU

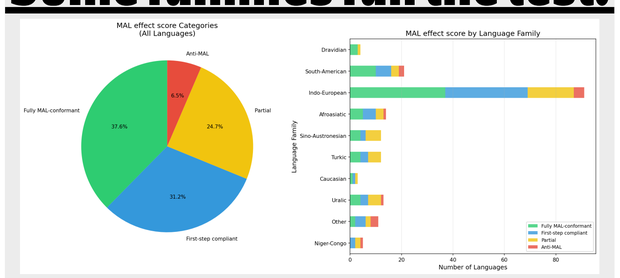
Left and right are not equal!



Left dependents (top) stay **flat** as n grows; right dependents (bottom) **collapse** from 5.2 down to 2.3. **Same law, two different worlds.** The verb is not ambidextrous — it leans right.

WAKE UP, INDO-EUROPEAN!

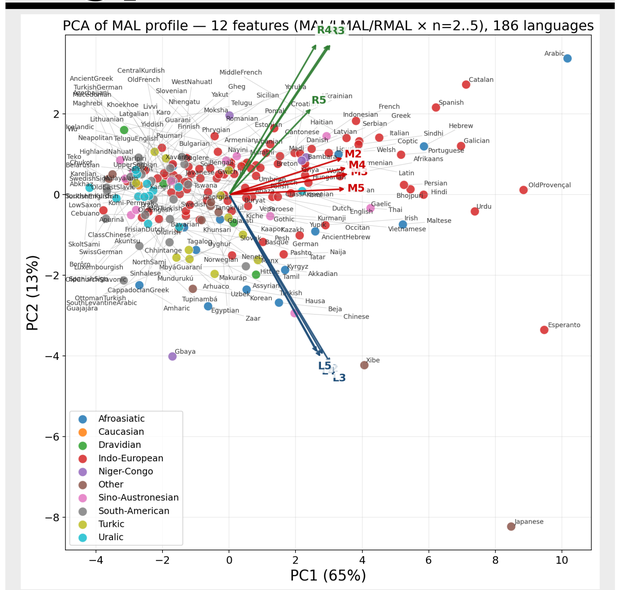
Some families fail the test!



Only 38% of languages are fully MAL-conformant. **6.5% are openly anti-MAL** — and every family hides at least one rebel. Indo-European is no model student.

THE DOTS THEY DON'T WANT YOU TO CONNECT

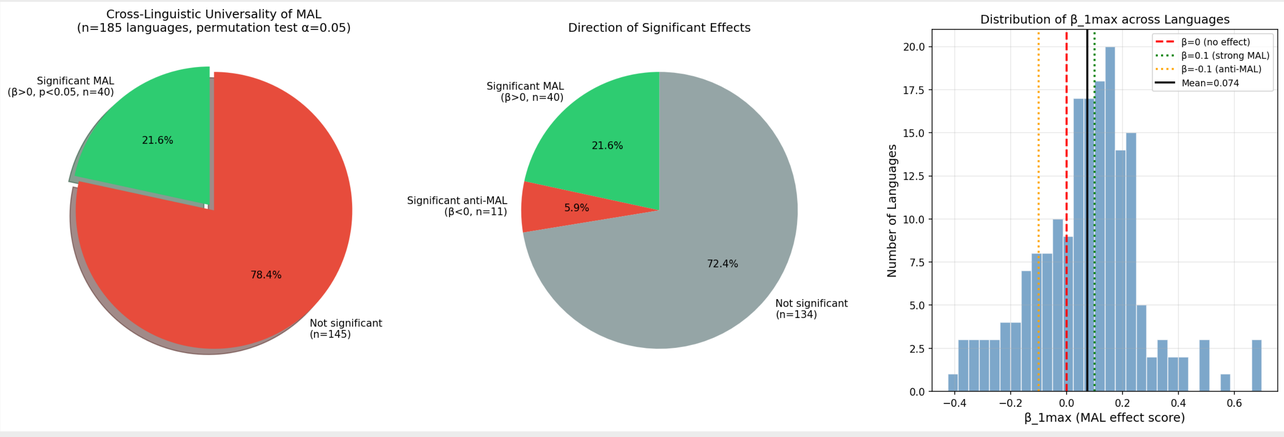
tiny points, BIG secrets!



Every language projected onto a 2D map of its MAL profile (PCA on 12 features: MAL/LMAL/RMAL at $n = 2..5$). **PC1 = 65%, PC2 = 13%.** Indo-European clusters tightly; Japanese, Esperanto and Old Provençal drift out as outliers.

THE 78% NOBODY TALKS ABOUT

Most languages flunk the test.

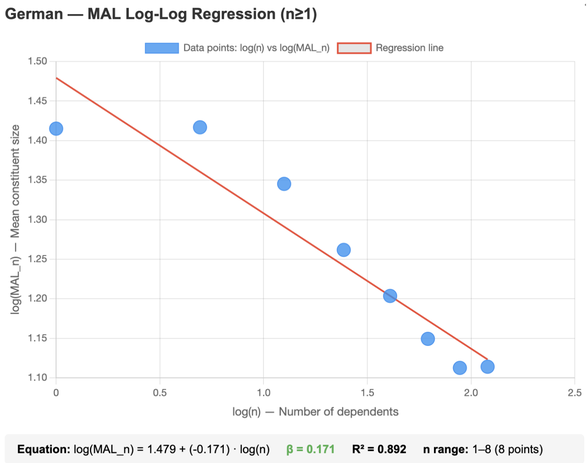


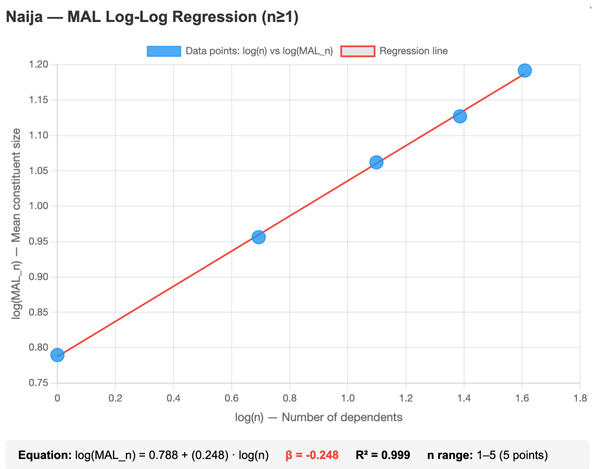
Only 21.6% of UD languages show a statistically significant MAL effect. **5.9%** go the *wrong* way (anti-MAL). The rest are flat: noise where the textbooks promised a law.

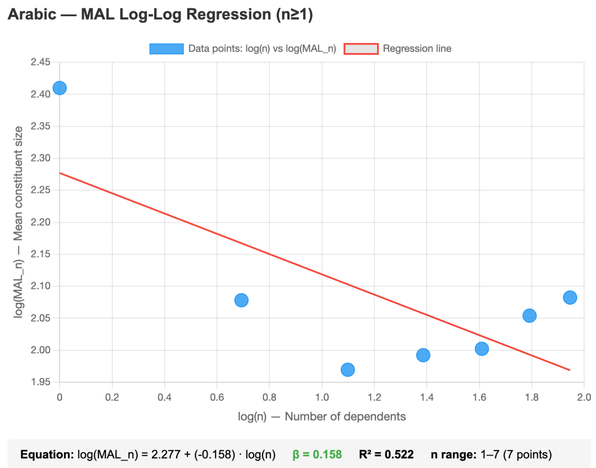
If MAL were a **linguistic universal**, this pie would be one big green slice. It isn't. Welcome to the post-universal era.

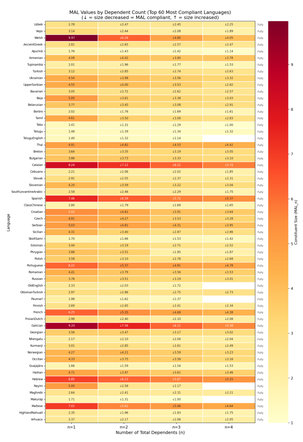
AS USUAL

German nails it. Naija wobbles. Arabic surprises. You?









German — textbook MAL

slope = -0.17, $R^2 \sim 0.99$ on the log-log fit. A near-perfect power law: every extra dependent shrinks the average constituent. **The cleanest fit in the sample** — not the strongest slope, but the most orderly one. Is anyone surprised?

Naija — mixed, asymmetric

Left and right dependents diverge wildly. A young pidgincreole refusing to fit one curve — in fact the **strongest Anti-LMAL in the whole sample** ($\beta = +0.73$ preverbally).

Arabic — near-textbook

Despite VSO order and rich morphology, Arabic lands close to German on the MAL map.

The wall of shame

Every row is a language, every column a transition $n \to n{+}1$. **Red = MAL holds, blue = MAL breaks down.** Mostly red — but the blue streaks tell you exactly where the law quietly fails.

IT'S 100% FREE*

Come and find out how your UD language is doing!

Scan, or come over to the laptop → typometrics.elizia.net/menzerath

180 treebanks, interactive PCA + family map + per-language curves + compliance ridgeline — all online. * *Free as in beer, free as in speech, free as in dependency. Side effects may include sudden curiosity about Sanskrit verb valency and the unsettling realisation that left and right are not the same.*

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